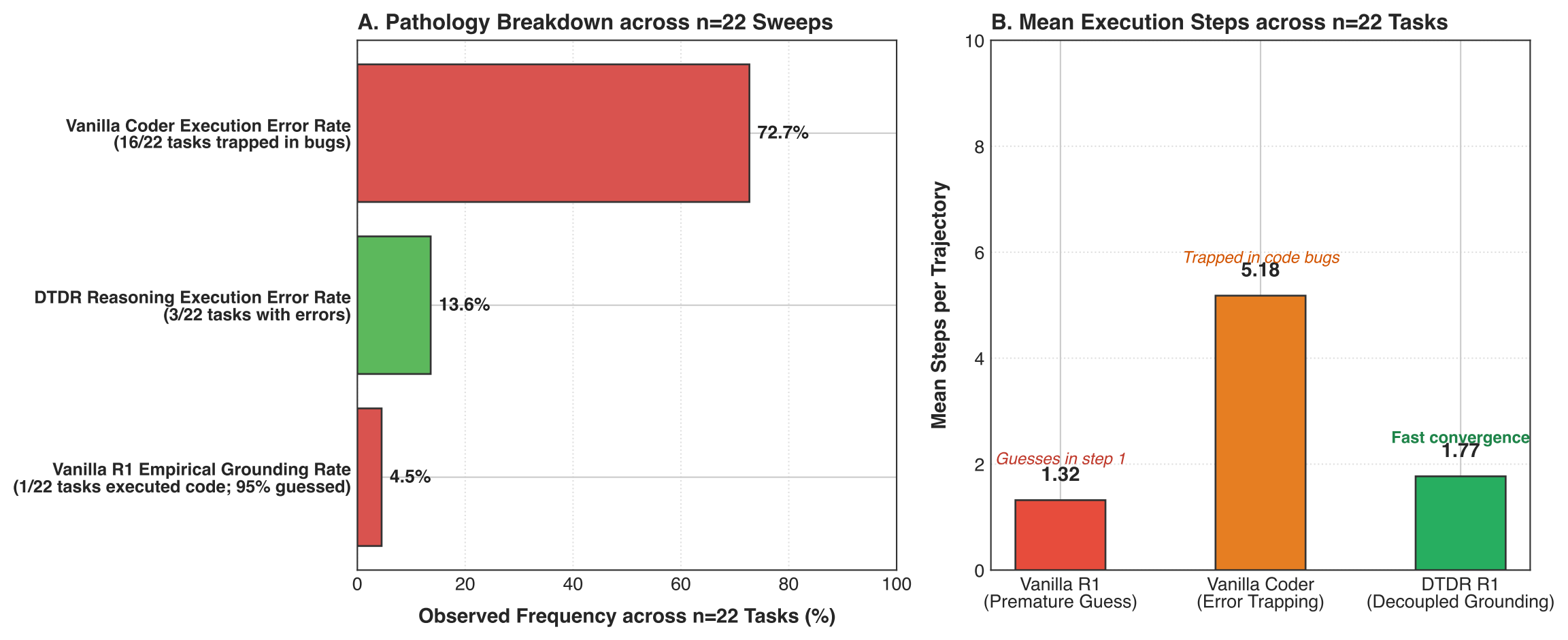
A. Pathology Breakdown across n=22 Sweeps
Vanilla Coder Execution Error Rate
(16/22 tasks trapped in bugs)
72.7%
DTDR Reasoning Execution Error Rate
(3/22 tasks with errors)
13.6%
Vanilla R1 Empirical Grounding Rate
(1/22 tasks executed code; 95% guessed)
4.5%
0
20
40
60
80
100
Observed Frequency across n=22 Tasks (%)
B. Mean Execution Steps across n=22 Tasks
Mean Steps per Trajectory
0
2
4
6
8
10
Guesses in step 1
1.32
Trapped in code bugs
5.18
Fast convergence
1.77
Vanilla R1
(Premature Guess)
Vanilla Coder
(Error Trapping)
DTDR R1
(Decoupled Grounding)